

Diabetes Classification

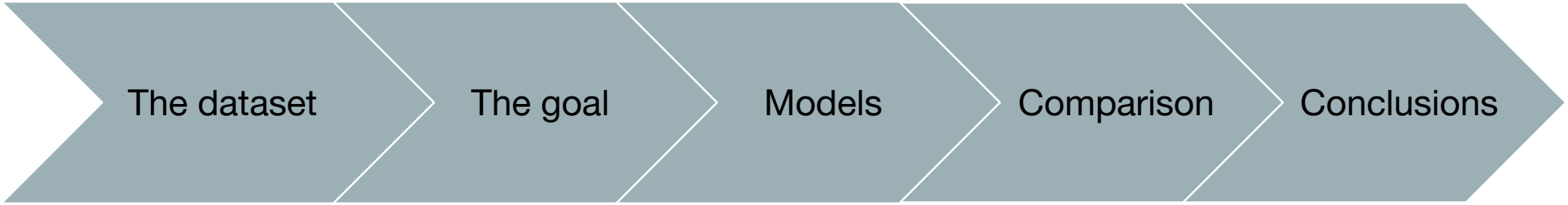
Statistical Learning – Data Analytics for Business

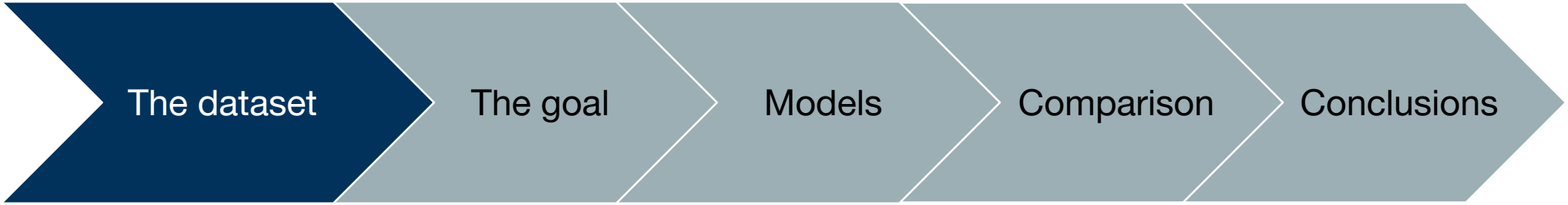
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Outline




The dataset
The goal
Models
Comparison
Conclusions

The dataset

The **Diabetes prediction dataset** is a collection of medical and demographic data from patients, along with their diabetes status (positive or negative).



- This dataset can be used to build machine learning models to predict diabetes in patients based on their medical history and demographic information.

- This can be useful for healthcare professionals in identifying patients who may be at risk of developing diabetes and in developing personalized treatment plans.

The dataset

Collection methodology



The dataset gathers medical and demographic data from patients who have been diagnosed with or are at risk of developing diabetes.

The data is typically collected through **surveys**, **medical records**, and **laboratory tests**.

→ Data source: *Electronic Health Records* (EHRs)

Data preparation



- Missing values detection → No missing values detected
- Presence of '*No Info*' for the `smoking_history` variable → '*No Info*' entries removed
- Categorical variables appropriately transformed into numerical representations

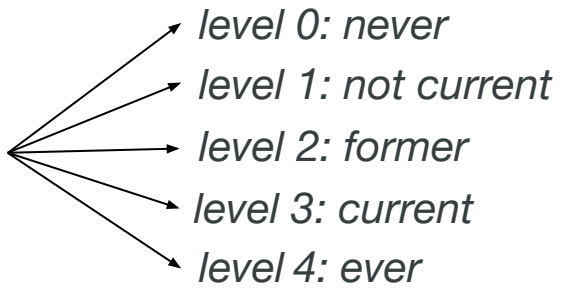
The dataset

100.000 observations and 9 variables:

- **`diabetes:`** binary variable
  - *level 1: presence of diabetes in the individual*
  - *level 0: absence of diabetes in the individual*
- **`hypertension:`** binary variable
  - *level 1: presence of hypertension in the individual*
  - *level 0: absence of hypertension in the individual*
- **`heart_disease:`** binary variable
  - *level 1: presence of heart disease in the individual*
  - *level 0: absence of heart disease in the individual*

The dataset

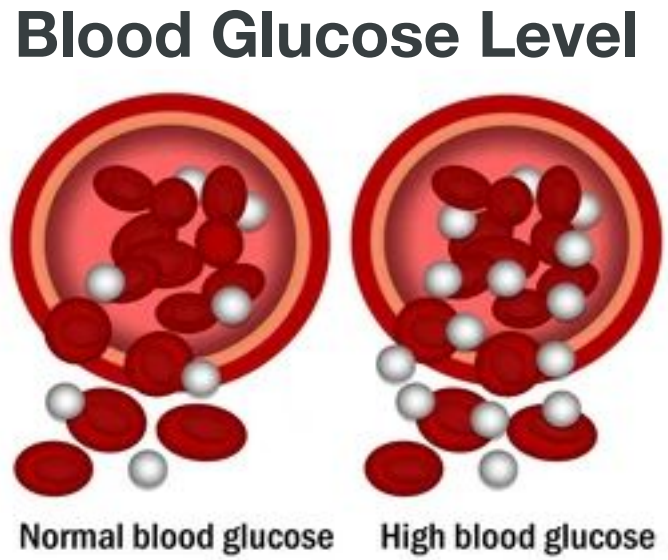
- **`smoking_history:`** categorical variable with 5 levels



- **`bmi:`** numerical variable

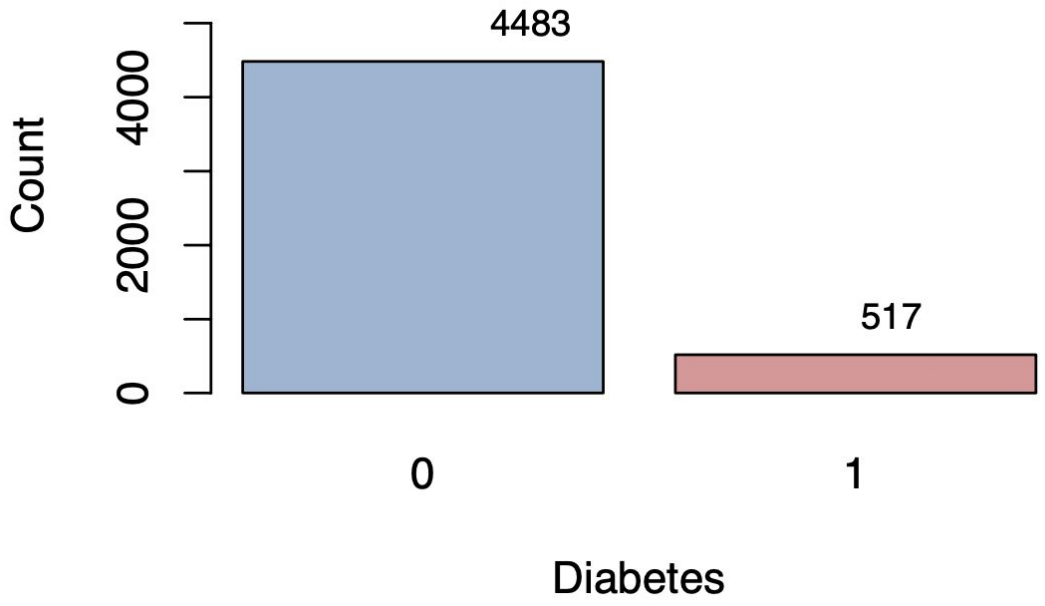
$$\begin{cases} \textit{underweight} & \text{if} \quad \textit{BMI} < 18.5 \\ \textit{normal} & \text{if} \quad 18.5 \leq \textit{BMI} < 25 \\ \textit{overweight} & \text{if} \quad 25 \leq \textit{BMI} < 30 \\ \textit{obese} & \text{if} \quad \textit{BMI} \geq 30 \end{cases}$$

- **`age:`** numerical variable
- **`gender:`** numerical variable
- **`blood_glucose_level:`** numerical variable
- **`Hba1c_level:`** numerical variable (Hemoglobine A1c level)

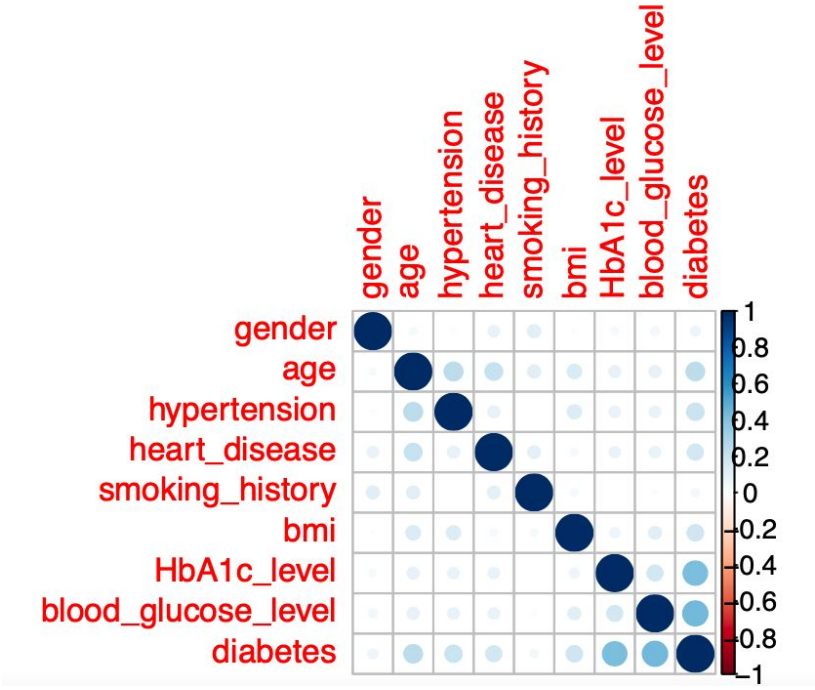


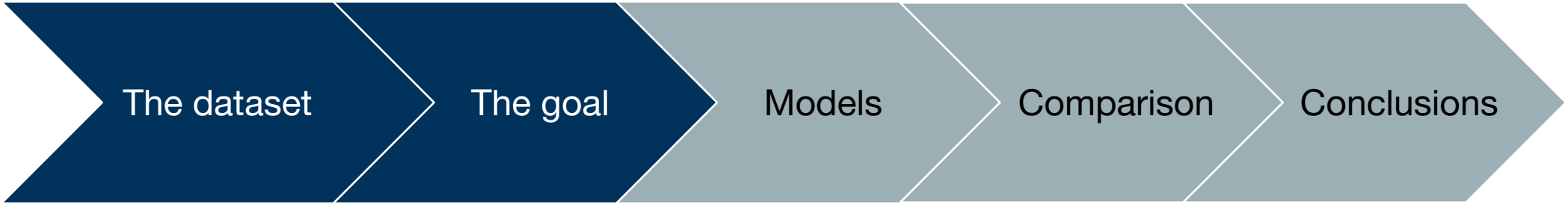
Response variable

Diabetes frequency distribution



Relationship among variables




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The goal

1. The goal of the project is to develop a classification model to distinguish between diabetic and non-diabetic patients using various statistical methods:

- *Logistic regression*
- *Support Vector Machines (SVM)*
- *K-Nearest Neighbors (KNN)*
- *Classification Trees*
- *Bagging*
- *Random Forest*

2. To make predictions on the likelihood of developing diabetes based on certain risk factors.


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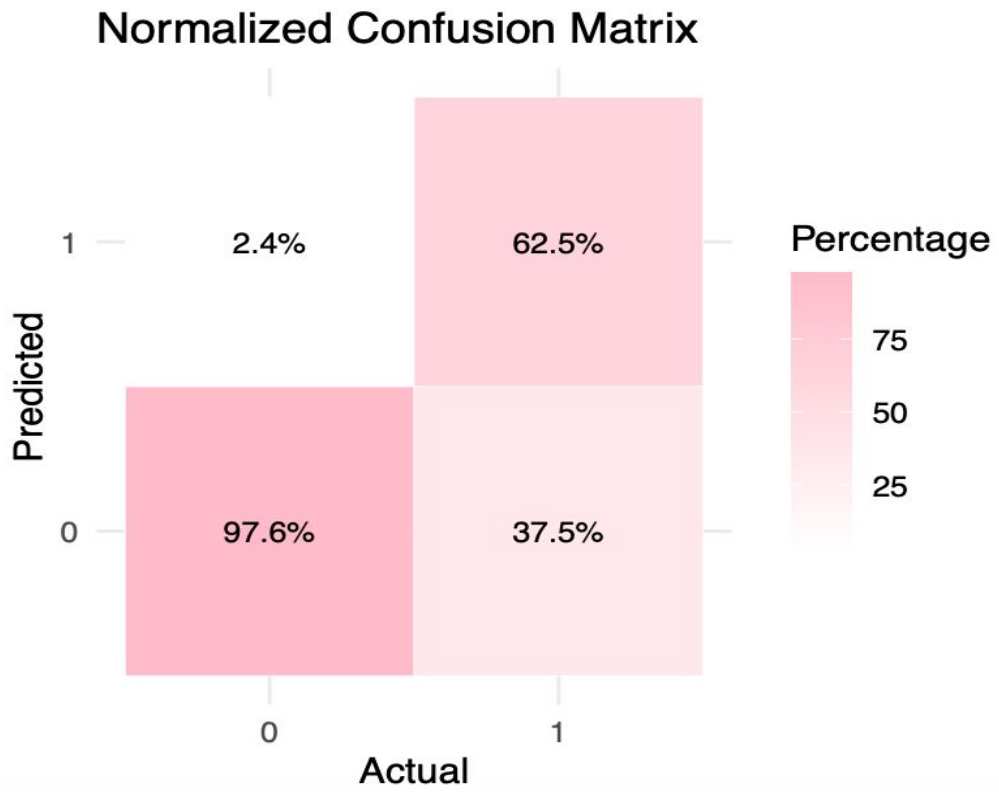
Logistic regression

- Generalized Linear Model (GLM) used for modeling binary outcomes by employing the *logit* function.
- Both categorical and numerical variables were included to fit this model.

Results

- The most influential variables include **`age`**, **`HbA1c_level`**, **`BMI`**, **`blood_glucose_level`** and **`hypertension`**.
- These findings align with established medical knowledge, where diabetes commonly occurs in older individuals with higher BMI and elevated blood glucose levels.

Logistic prediction



Accuracy	0.942
Specificity	0.9757
Sensitivity	0.6250

Support Vector Machines (SVM)

Support Vector Machines (SVM) is an approach for binary classification.

SVM aims to find the optimal hyperplane that separates the two classes within the feature space, while maximizing the margin. The margin refers to the distance between the hyperplane and the closest data points from each class, known as support vectors.

By maximizing the margin, SVMs aims for:
- **accurate predictions** on the training data
- **robust generalization** to unseen data.

Notes:
- *SVM can accomodate non-linear decision boundaries through the use of the **radial kernel** function.*
- *SVM exclusively operates with **numerical variables**.*
- *SVM can be very **sensible to different scale of measurements**.*

Parameters Tuning

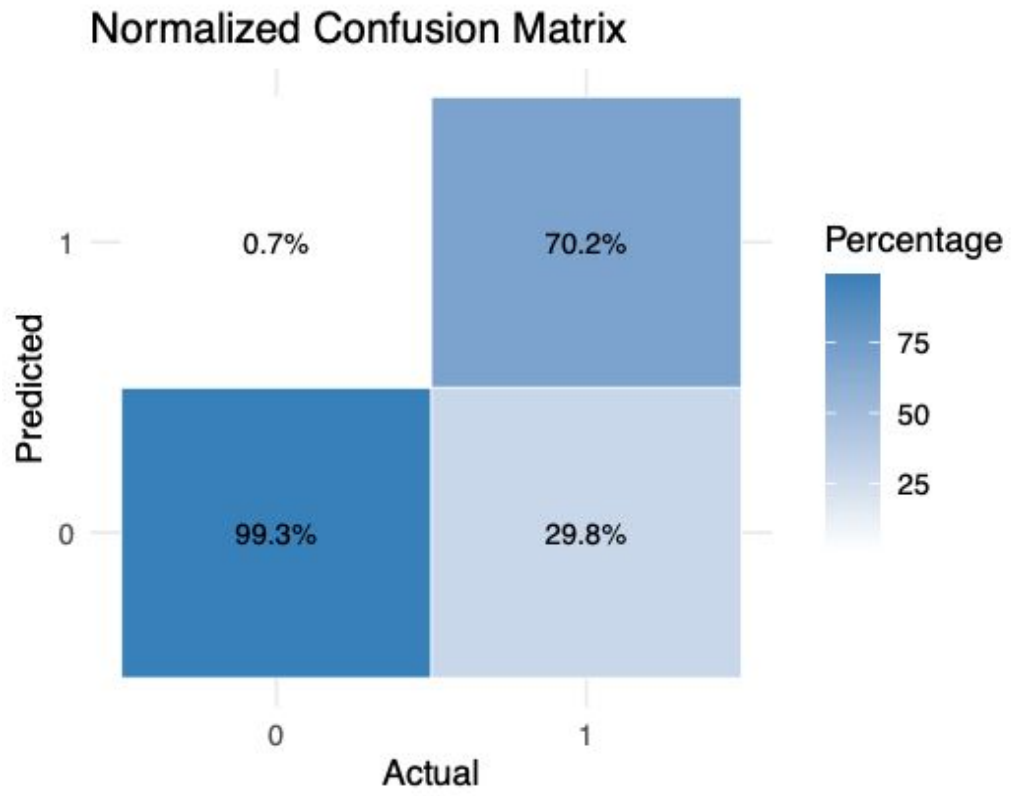
The model was trained with a radial kernel.
It was performed a 10 fold Cross Validation to select the optimal values of the parameters cost and gamma

CV results

- **cost** = 10
- **gamma** = 1

cost and **gamma** are used to control the **trade-off between model complexity and the ability to fit the training data**.

SVM prediction



Accuracy	0.966
Specificity	0.9934
Sensitivity	0.7021

K-Nearest Neighbors (KNN)

The K-Nearest Neighbors (KNN) is an approach for classification and regression.
It is a non-parametric method that makes predictions based on the data included in neighborhoods of each point.
The idea behind is that **similar data points tend to have similar labels**. It relies on the fact that there exists a distance between covariates.

It estimates the distribution of Y|X and it uses the estimated probability to classify data.

$$\hat{P}[Y = c_j | X = x_i] = \frac{1}{K} \sum_{i \in N_i^K} I(y_i = c_j)$$

where:
- $\hat{P}[Y = c_j | X = x_i]$ is the posterior probability of class *j*
- N_i is the neighborhood of x_i : K points closest to x_i (with respect to a distance d)

Notes:
- *KNN exclusively operates with **numerical variables**.*
- *KNN can be very **sensible to different scale of measurements**.*

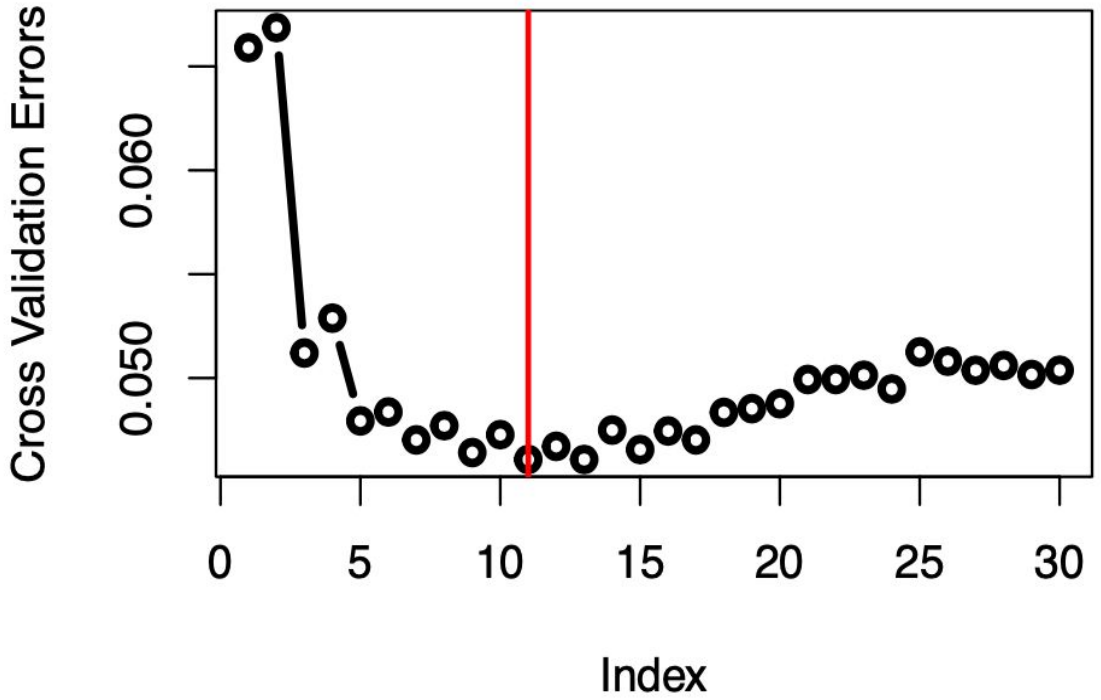
Cross-Validation

Cross-Validation (CV)

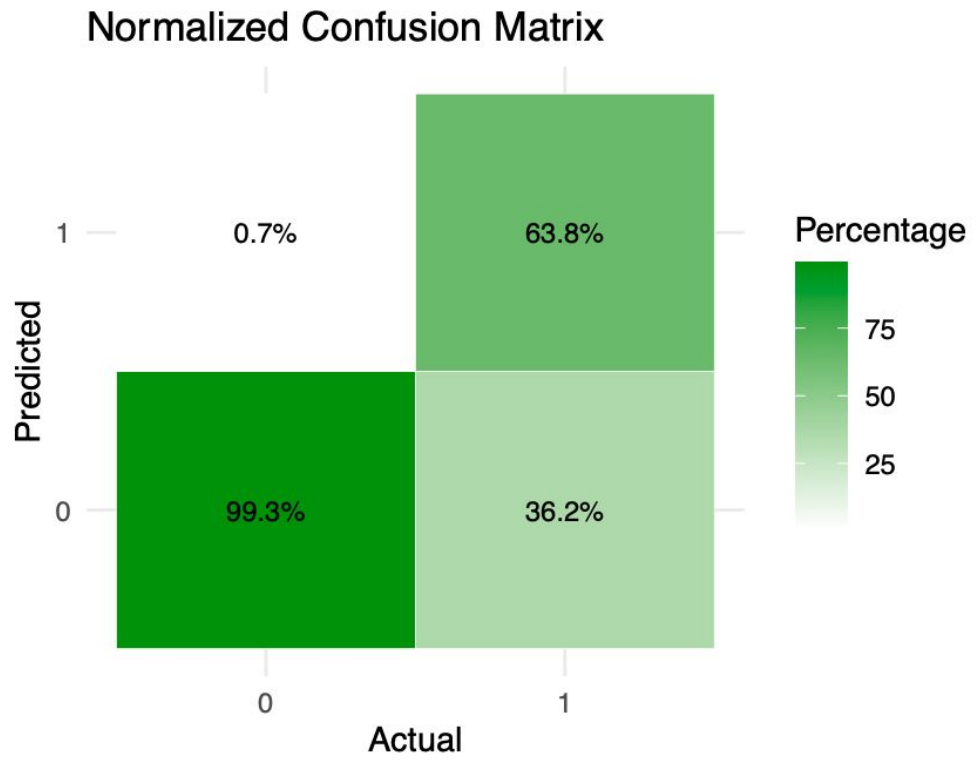
The best k selected through 10-fold Cross Validation approach

CV results

The number of neighbors selected is k=11



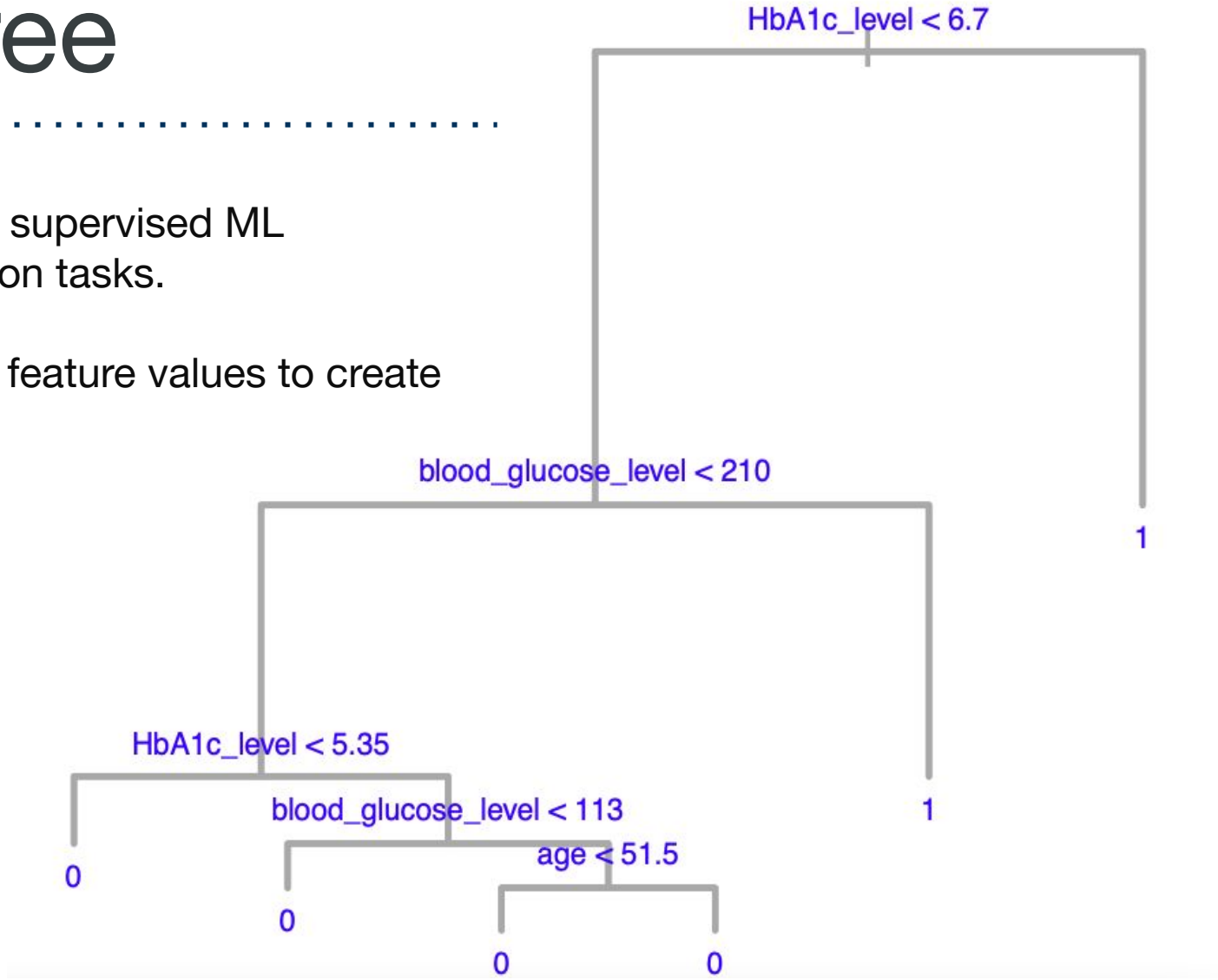
KNN Prediction



Accuracy	0.96
Specificity	0.9934
Sensitivity	0.6383

Classification Tree

- Classification tree, or decision tree, is a supervised ML algorithm for classification and regression tasks.
- It partitions data into subsets based on feature values to create predictive models.
- Splitting guided by metrics like Gini impurity or information gain, assessing class homogeneity.
- Algorithm continues splitting until a stopping criterion, e.g., maximum depth or minimum samples, is met.



Tree Pruning

Why pruning the tree?

Recursive binary splitting the data is likely to overfit data.

How pruning works?

Through Classification Error Rate which measures the proportion of misclassified training observations.

$$E_{R_m} = 1 - \max_k(\widehat{p}_{mk})$$

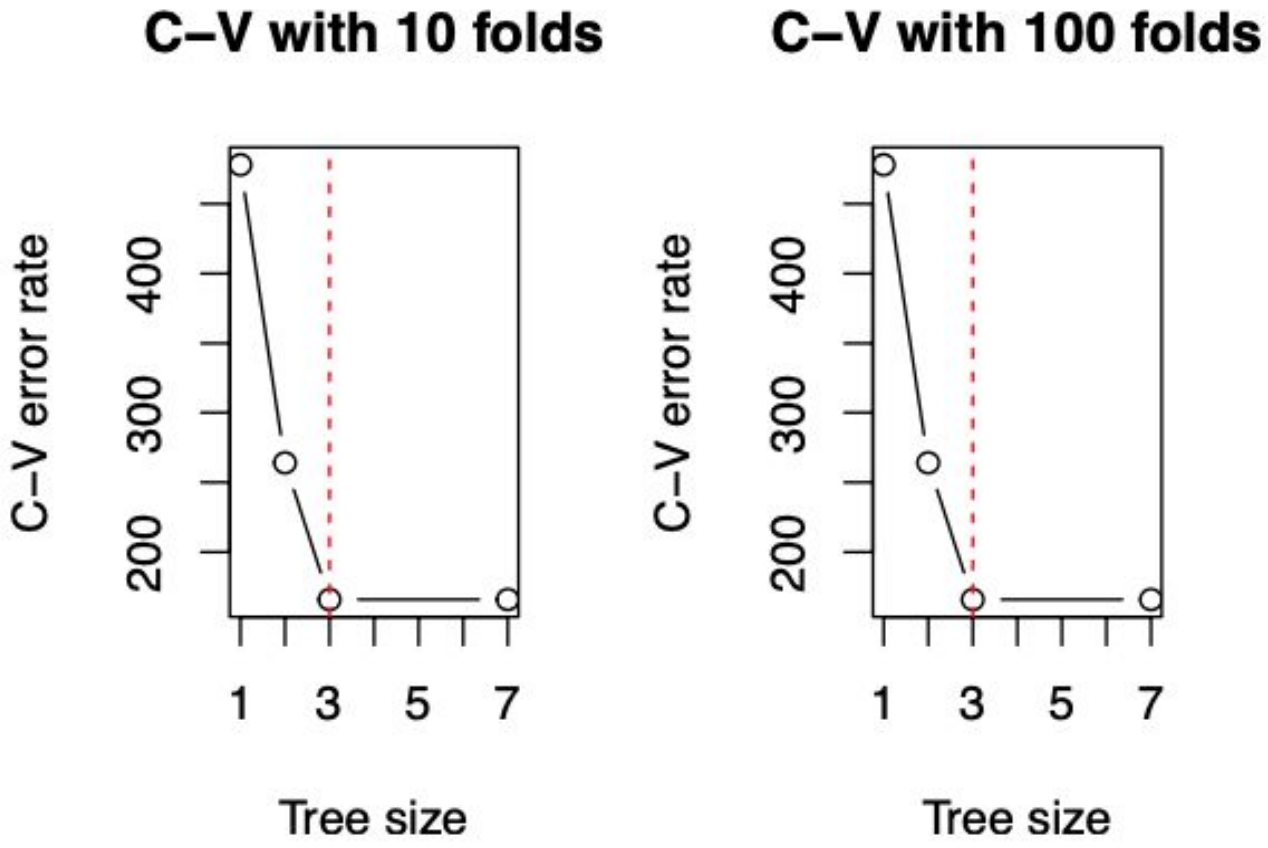
Pruning and Cross-Validation (CV)

Cross-Validation (CV)

The optimal size of the tree is obtained through 10 folds and 100 folds Cross Validation

CV results

The optimal Tree size is 3 terminal nodes.



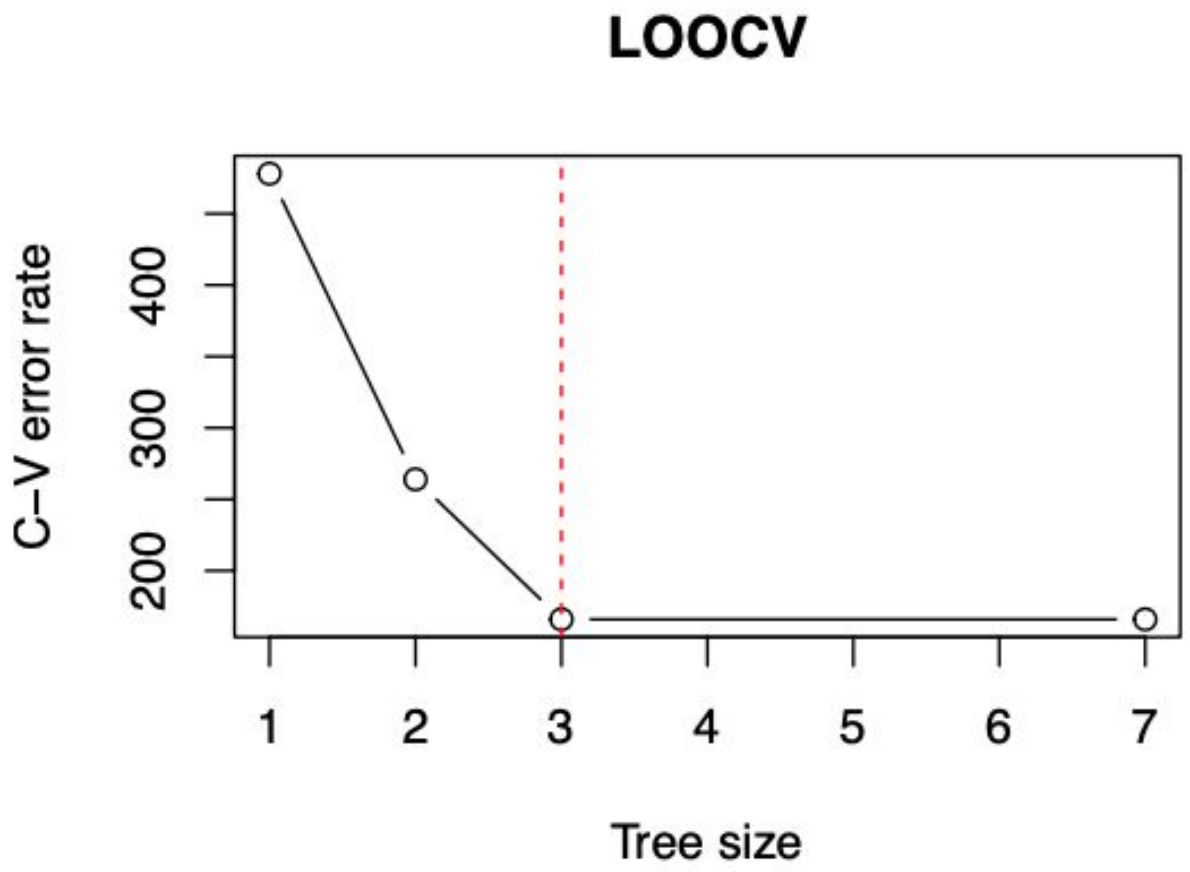
Leave-One-Out-Cross-Validation (LOOCV)

Leave-One-Cut-Cross-Validation (LOOCV)

LOOCV is used to validate the previous results in terms of optimal size of the tree

LOOCV results

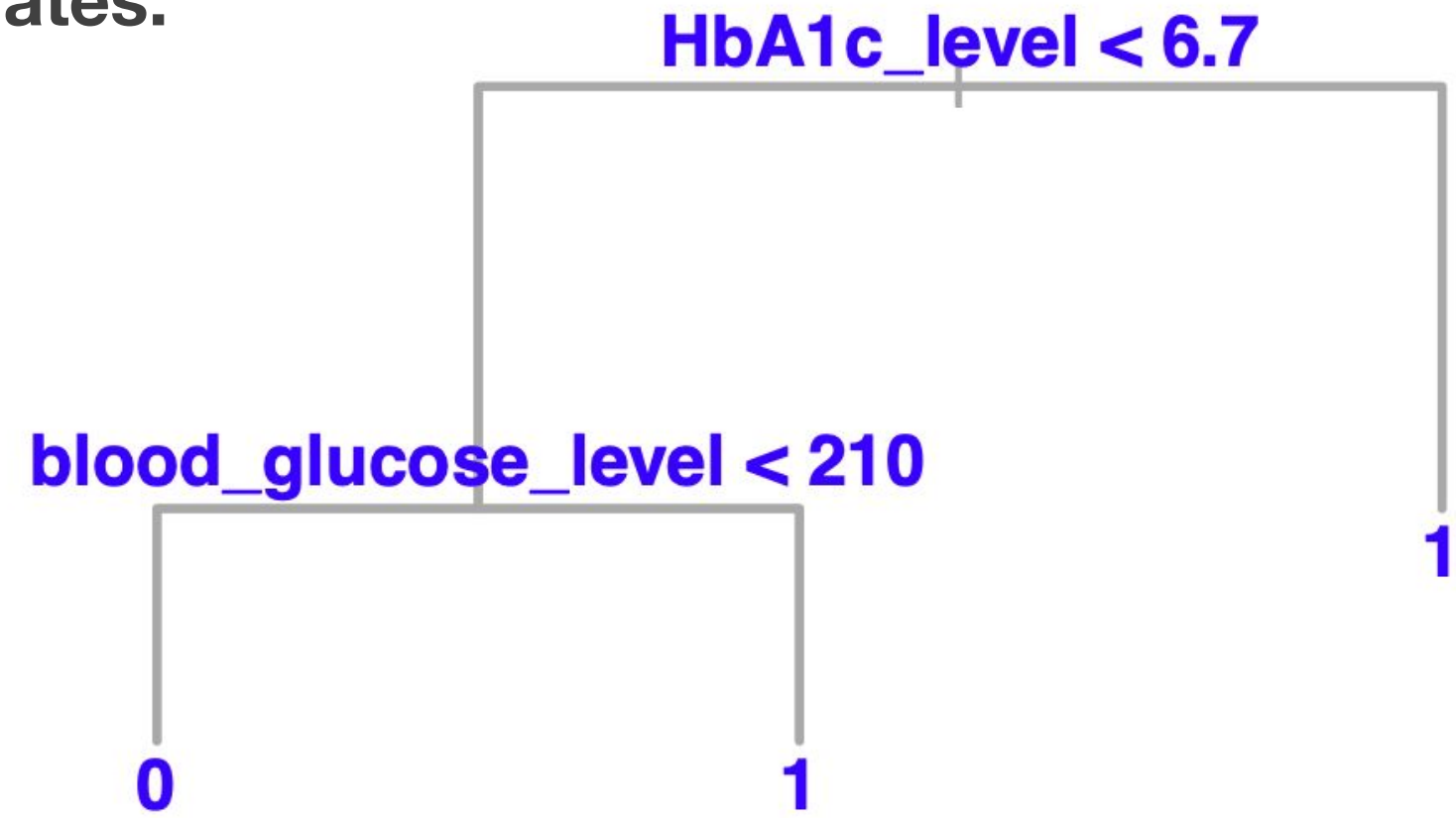
The optimal Tree size is 3 terminal nodes.



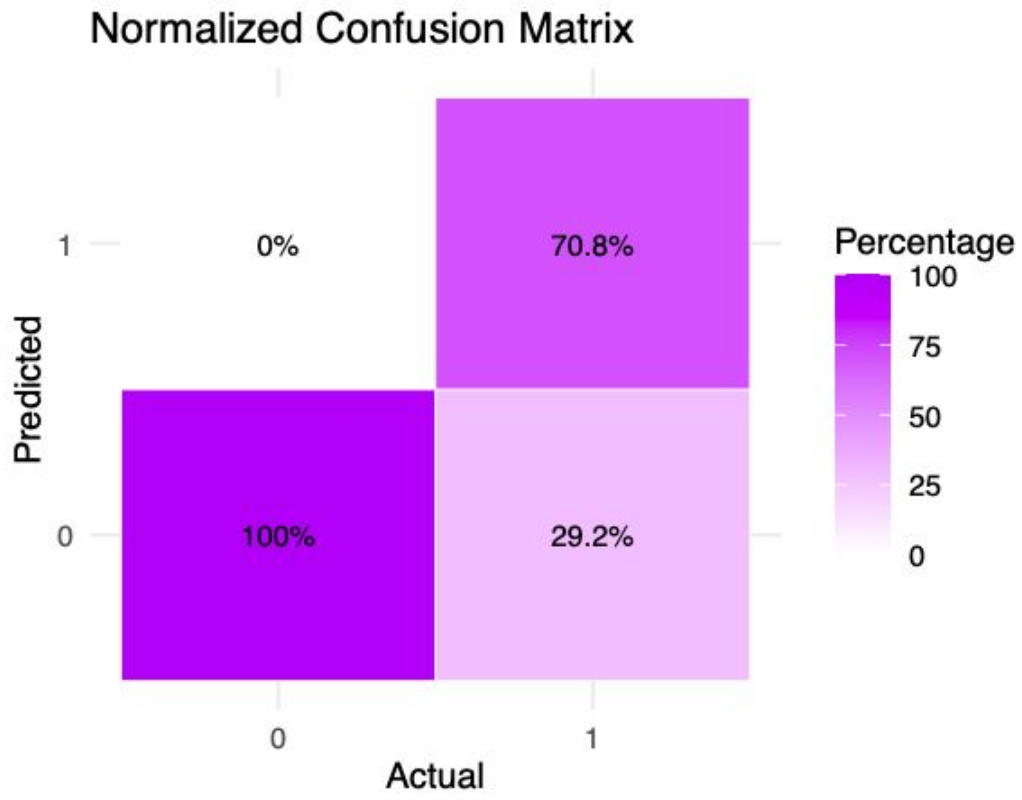
Pruned Tree

Most informative covariates:

- `HbA1c_level`
- `blood_glucose_level`



Pruned Tree prediction



Accuracy	0.972
Specificity	1
Sensitivity	0.7083

Bagging

A method that applies ideas from the **Bootstrap** to estimate more stable trees

- **Boostrap Sampling:** generate *B* Bootstrap samples by randomly sampling *N* observations with replacement from the original dataset.
- **Model Fitting:** fit a base model on each *B*
- **Aggregation:** combine the predictions from all base models and choose the one with the highest sum of predictions.

Model Fitting

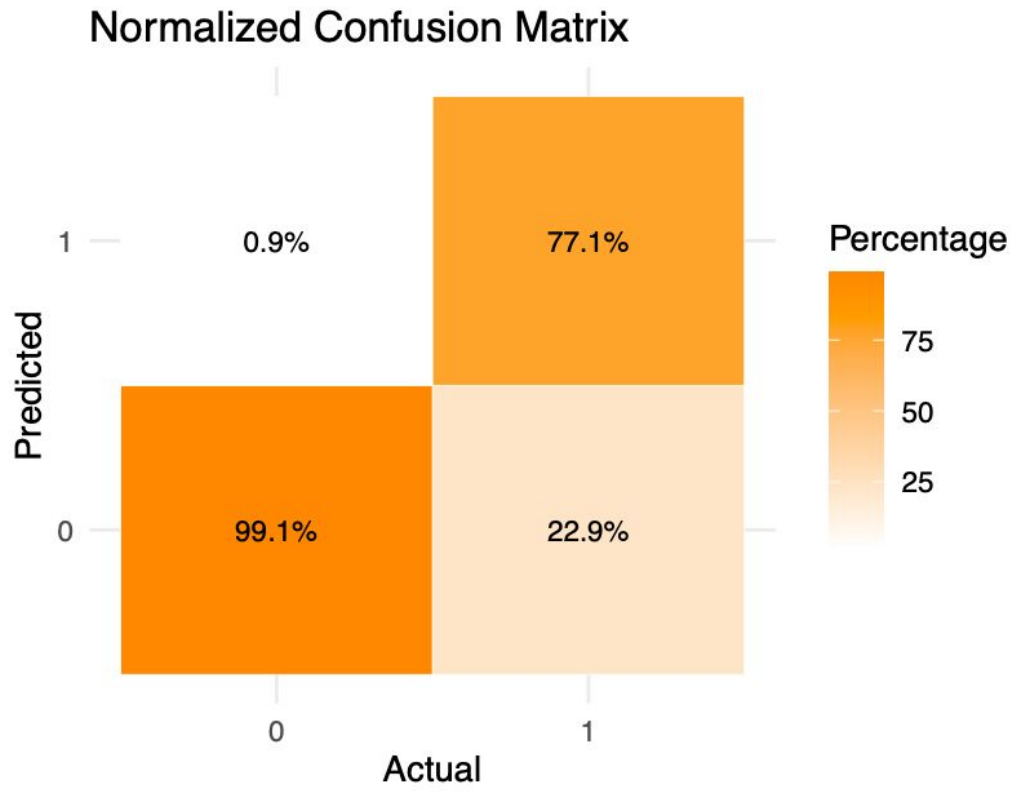
randomForest( ) function with the following parameters:

- `mtry = 8`
- `B = 1000`

Results

OOB error rate: 3,64% → strong performance

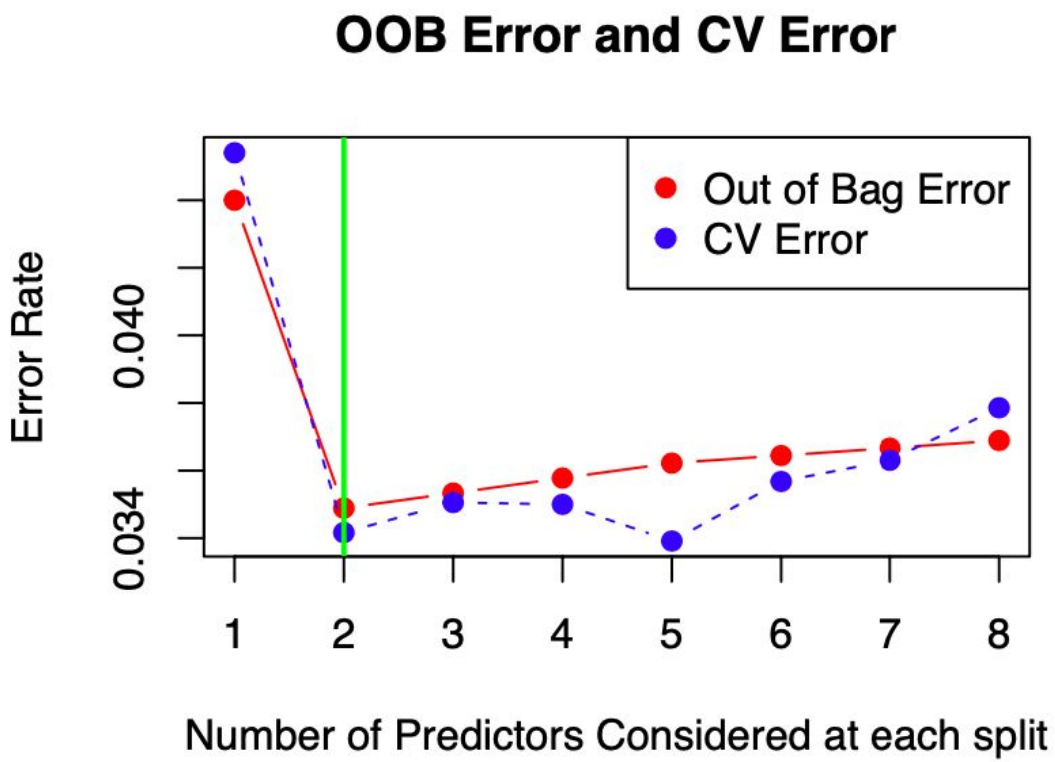
Bagging prediction



Accuracy	0.97
Specifcity	0.9912
Sensitivity	0.7708

Random Forest

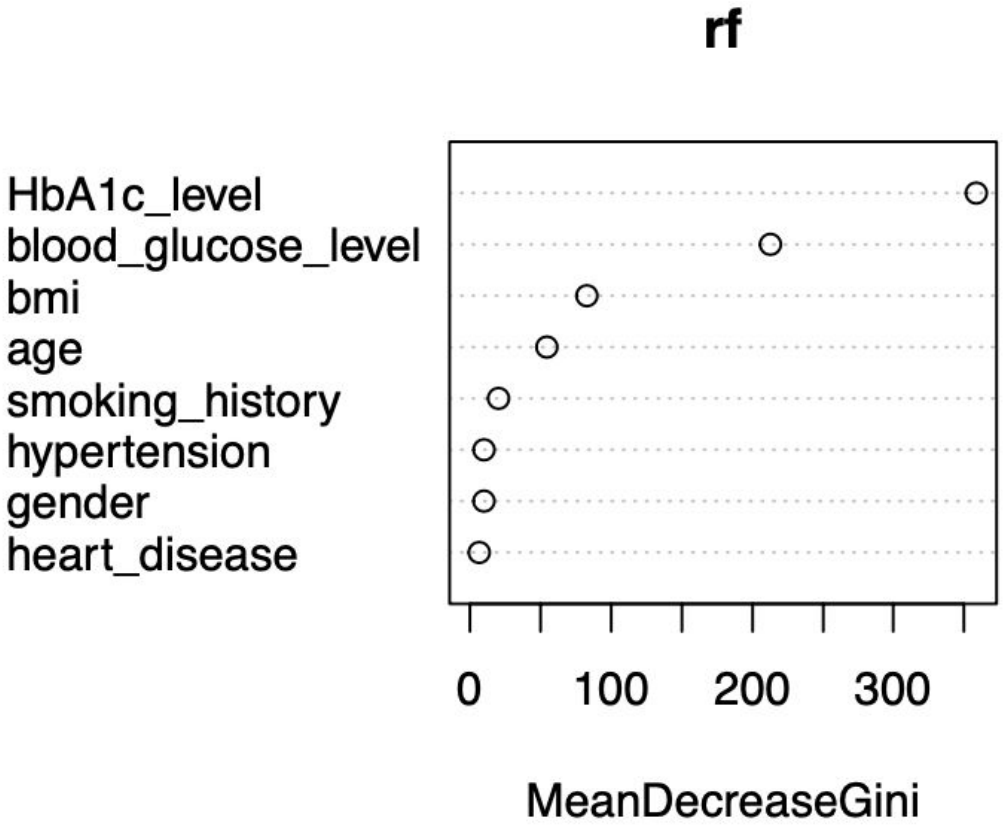
`mtry` exploration through the *OOB error* and *CV error*



- A noticeable decline in error, followed by a subsequent increase, occurs at the level of the **second variable**.

Exploration result ⟹ **`mtry = 2`**

Random Forest



Most influential variables:

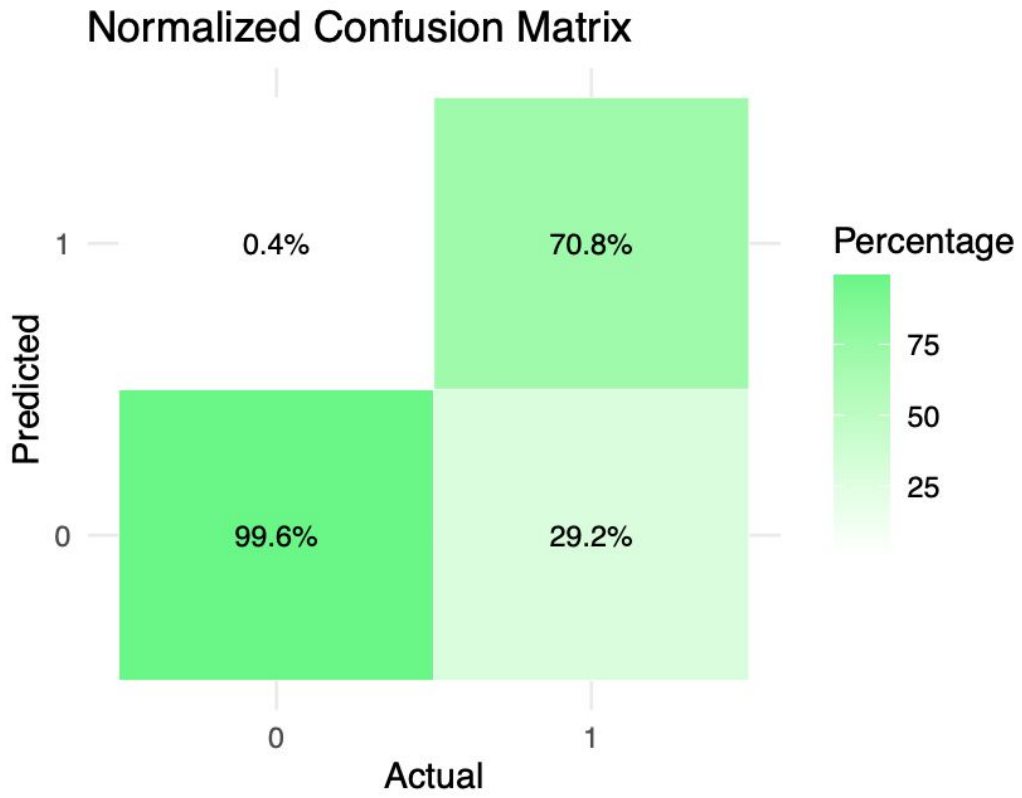
- **`HbA1c_level`**
- **`Blood_glucose_level`**

Model Fitting

randomForest( ) function with the following parameters:

- **`mtry = 2`**
- **`B = 1000`**

Random forest prediction



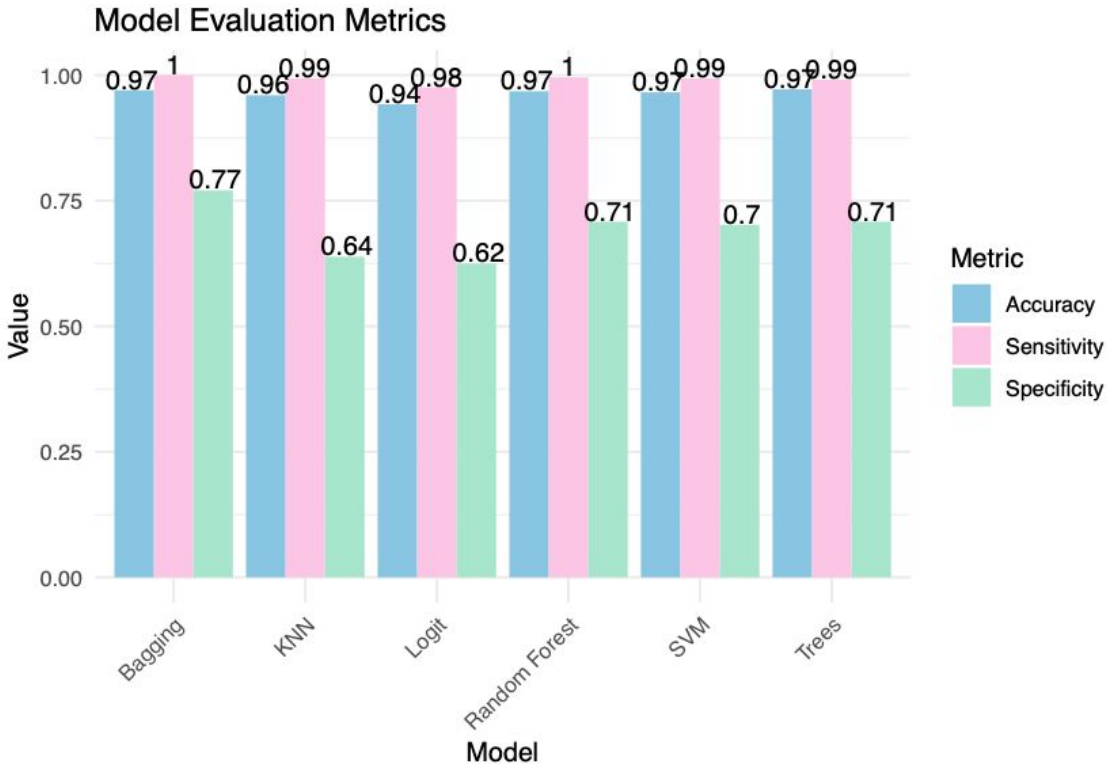
Accuracy	0.968
Specificity	0.9956
Sensitivity	0.7083


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Sensitivity

Model Comparison

Model	Accuracy	Sensitivity	Specificity
Logit	0.942	0.9757	0.6250
SVM	0.966	0.9934	0.7021
KNN	0.960	0.9934	0.6383
Trees	0.972	0.9912	0.7083
Bagging	0.970	1.0000	0.7708
Random Forest	0.968	0.9956	0.7083




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Conclusions

- All six models displayed consistent performance in accuracy, sensitivity, and specificity, indicating robustness in predicting diabetes.
- Despite satisfactory overall accuracy, sensitivity scores revealed challenges in accurately predicting instances of diabetic individuals, possibly due to dataset imbalance with fewer diabetic observations.
- Bagging, Trees and SVM models emerged as promising options, showcasing commendable accuracy, specificity, and competitive sensitivity scores.
- Further exploration and refinement of these models could improve predictive capabilities for diabetic classification, but also performing a data augmentation for the category “diabetes” of the response variable.

Thank you

Lorenzo Bonatti Sara Cassiani Chiara Mensini